

A Dual-Policy Framework for Safe and Continuous Humanoid Locomotion in Dynamic Construction Environments

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Abstract

The deployment of humanoid robots in dynamic environments, such as construction sites, presents a significant challenge in robotics: enabling continuous locomotion while simultaneously avoiding upper-body collisions. Existing control frameworks often treat walking and obstacle avoidance as separate, sequential tasks, leading to inefficient stop-and-go behaviors or potential collisions. This paper presents a dual-policy control framework that addresses this issue by decoupling the whole-body control problem. The framework integrates two specialized policies: (1) a lower-body locomotion policy trained using deep reinforcement learning for robust and adaptive walking, and (2) an upper-body reactive safety policy based on Control Barrier Functions (CBFs) that provides formal guarantees for collision avoidance. A lightweight coordination layer ensures that upper-body motions do not compromise the robot’s dynamic stability. We validate our framework in a high-fidelity simulated construction environment, demonstrating that this decoupled approach achieves continuous, safe, and efficient navigation through scenarios with both static and dynamic obstacles, outperforming traditional monolithic control paradigms.

Keywords:

1. Introduction

Humanoid robots are increasingly recognized for their potential to operate in complex, human-centric environments, with applications ranging from logistics to disaster response [1, 2]. Their anthropomorphic design allows

them to navigate spaces and use tools intended for humans, making them particularly suitable for unstructured settings like construction sites [3, 4]. As these platforms become more capable, the focus shifts from basic locomotion to the integration of more complex skills, such as performing tasks while in motion [5, 6].

A fundamental capability for any mobile robot in a dynamic environment is the ability to walk and react to its surroundings simultaneously, a challenge often referred to as loco-manipulation or loco-avoidance [1, 7]. Many current systems, however, are unable to perform these actions concurrently. They often resort to inefficient "stop-and-go" strategies, where the robot must halt its locomotion to perform an upper-body action, such as avoiding an obstacle or manipulating an object [1]. This sequential approach is impractical in cluttered and dynamic spaces where continuous movement is essential for efficiency and safety [8, 9].

This work addresses the specific problem of enabling a humanoid robot to walk continuously while its upper body reactively avoids obstacles. This capability is critical for navigating the confined and unpredictable pathways of a construction site. Existing whole-body control paradigms present limitations in this context. Monolithic, optimization-based controllers that solve for all joints simultaneously, such as those combining Model Predictive Control (MPC) with a Whole-Body Controller (WBC), can generate optimal motions but often struggle with the computational demands required for real-time reactivity to unforeseen obstacles [10, 11, 12].

Conversely, the field of learning-based locomotion has produced remarkably robust walking policies using deep reinforcement learning (RL) [13, 14, 15]. These policies can traverse challenging and varied terrains, a key requirement for construction sites [3, 16, 17]. However, to manage the complexity of the learning problem, many state-of-the-art locomotion controllers treat the upper body passively, using it primarily for balance or constraining it to a fixed posture [18, 1]. These policies are not explicitly trained to use the torso, arms, and head for active, goal-directed obstacle avoidance while walking. This creates a critical gap between the demonstrated walking robustness and the practical need for whole-body environmental interaction.

To bridge this gap, this paper presents a novel, decoupled dual-policy architecture. Our approach divides the high-dimensional control problem, assigning specialized controllers to the upper and lower body.

- **Lower-Body Locomotion Policy:** A robust walking policy, trained

using deep reinforcement learning, controls the legs and waist. This policy is optimized to maintain stable bipedal locomotion over uneven terrain, leveraging recent advances that have demonstrated exceptional adaptability [3, 19, 20].

- **Upper-Body Safety Policy:** A reactive avoidance policy, formulated as a safety filter using Control Barrier Functions (CBFs), controls the torso, arms, and head. This approach provides mathematically-grounded, provable guarantees of collision avoidance, ensuring the robot’s upper body makes the minimal necessary adjustment to remain safe without halting its forward progress [9, 21, 22].

A lightweight coordination layer dynamically links the two policies, ensuring that reactive upper-body motions do not exceed the stability limits of the current walking gait. This synergistic combination of learning-based robustness and formal safety provides a practical and effective solution for continuous navigation in cluttered environments. Our primary contribution is the design and validation of this dual-policy architecture, which demonstrates superior performance in achieving safe and fluid navigation compared to monolithic or purely reactive approaches.

2. Related Work

Research in humanoid robotics has long focused on Whole-Body Control (WBC) to coordinate the many degrees of freedom required for complex tasks [23, 24]. A dominant paradigm combines high-level Model Predictive Control (MPC) with low-level optimization-based controllers to generate dynamically consistent motions [10, 11, 12]. While these model-based methods are powerful, their computational complexity can present challenges for real-time reactivity to unforeseen events. In parallel, the field of bipedal locomotion has been significantly advanced by deep reinforcement learning (RL), which has produced remarkably robust walking policies over varied and challenging terrains without extensive manual engineering [25, 13, 15]. This approach has shown particular promise for navigating the unstructured surfaces found on construction sites [3, 14, 26, 17].

A common approach in many successful RL locomotion frameworks is to simplify the learning problem by treating the upper body passively, using it primarily for balance rather than active tasks [1]. This observation has led to a promising architectural trend: decoupling the control problem. Recent

work, for instance, has successfully combined an RL-based locomotion policy with inverse kinematics (IK) for precise upper-body manipulation [7]. Other frameworks have used a similar division to enable expressive upper-body motions during walking, drawing from human motion datasets [18]. Our work is philosophically aligned with this decoupled approach. However, instead of focusing on precise manipulation or expressive imitation, we address the critical need for reactive safety, employing a formal safety framework for the upper body.

Ensuring safety is a non-negotiable requirement for robots in human-centric environments [4]. To provide rigorous safety assurances, the field has increasingly adopted formal methods like Control Barrier Functions (CBFs), which define a "safe set" of states and synthesize a controller guaranteed to remain within it [21]. This paradigm is well-suited for the "dexterous safety" problem in cluttered environments, as explored by Chen et al., where numerous limb-level collision constraints must be managed simultaneously [8]. The importance of such modular and verifiable safety layers is further evidenced by the development of standardized benchmarks like SPARK [9]. Our framework leverages this mature safety paradigm for the humanoid upper body, providing a level of verifiable safety that is difficult to achieve with purely heuristic methods. By integrating this formal safety component with a learned locomotion policy, we aim to bridge a specific gap noted in construction robotics literature, where dynamic locomotion has been achieved but often without simultaneous, reactive upper-body collision avoidance [3].

3. Methodology

3.1. System Architecture Overview

3.2. Lower-Body Locomotion Policy

3.3. Upper-Body Obstacle Avoidance Policy

3.4. Coordination Mechanism

4. Results and Analysis

Results and Discussion...

5. Conclusion and Future Work

Conclusion...

Appendix A. Example Appendix Section

Appendix text.

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